

Question No:9

Problem Description

There are N chocolates denoted by array A where $A[i]$ is the length of the i -th chocolate.

Alice can melt each chocolate and then convert it into a chocolate whose length is any divisor of the number $A[i]$.

So, a chocolate of length $A[i]$ can be converted into X different types of chocolate where X is the count of divisors of the number $A[i]$.

So you need to count the total unordered pair of chocolates such that their X value is same.

Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq A[i] \leq 10^6$$

Input Format

The first line contains an integer N as input denoting the total number of elements in the array A .

The next line contains N space-separated integers that denote the elements of the array A .

Output Format

In the output, print the total number of ways as mentioned in the statement.

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
3	10
2 3 4	2 3 4 2 7 6 8 6 4 9
EXPECTED OUTPUT	EXPECTED OUTPUT
1	12

Code:

```

1  #include <stdio.h>
2
3  #define MAX 1000000
4
5  int divisors[MAX + 1];
6  int freq[MAX + 1];
7  int A[100005];
8
9  int main() {
10     int N;
11     scanf("%d", &N);
12
13     int max = 0;
14
15     for (int i = 0; i < N; i++) {
16         scanf("%d", &A[i]);
17
18         if (A[i] > max) {
19             max = A[i];
20         }
21     }
22
23     // Count divisors of each number
24     for (int i = 1; i <= max; i++) {
25         for (int j = i; j <= max; j += i) {
26             divisors[j]++;
27         }
28     }
29
30     // Count how many chocolates have each divisor count
31     for (int i = 0; i < N; i++) {
32         freq[divisors[A[i]]]++;
33     }
34
35     // Calculate the number of matching pairs
36     long long ans = 0;
37
38     for (int i = 1; i <= max; i++) {
39         ans += (long long)freq[i] * (freq[i] - 1) / 2;
40     }
41
42     printf("%lld\n", ans);
43
44     return 0;
45 }

```

Output:

Test Case-1:

```
3sem\DSA\DSA Practice\H  
3  
2 3 4  
1  
PS C:\Users\kesha\OneDr
```

Test Case-2:

```
10  
2 3 4 2 7 6 8 6 4 9  
12
```

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